

Assessment Proforma 2025-26

Key Information

Module Code	CMT224 and CM4224
Module Title	Social Computing
Module Leader	Dr Liam Turner
Module Moderator	Professor Omer Rana
Assessment Title	Social Computing Problem Sheets
Assessment Number	1
Assessment Weighting	100%
Assessment Limits	Submissions must be made using the notebook templates provided. Follow the instructions under “Assessment Description” below.

This assessment's due date and time is specified on Learning Central. To view it click on the assessment's link, which will open '*Details & Information*' section for this assessment. Alternatively, go to '*Calendar*' section of a module, and select '*Due Dates*' option. You can change the view to daily or monthly, as well as use arrows to navigate the calendar.

The COMSC Assessment Calendar for all assessments is also available on Learning Central: by going to '[COMSC-SCHOOL](#)' organisation -> '**Assessment & Feedback 2025 - 26**' folder -> '[COMSC Assessment Calendar 2025 - 26](#)' page.

Learning Outcomes

The learning outcomes for this assessment are as follows:

1. Analyse fundamental traits of complex networks by synthesising theoretical concepts and methodologies from graph theory.
2. Evaluate and implement computational approaches to model and visualise complex social phenomena.
3. Design and create software to investigate or support human interaction behaviour.

Submission Instructions

All assessment files should be submitted in 'Assessment & Feedback' folder on Learning Central.

Before you submit your assessment, you **must complete and submit** 'Assessment Coversheet' form. Further instructions are given in this module's 'Assessment & Feedback' folder.

Your submission for this assessment should contain the following file(s):

Description	Compulsory or Optional?	Type	Name
Coversheet	Compulsory	One PDF (.pdf) file	[student_number]-coversheet.pdf
Part 1 Notebook (Using the template provided on Learning Central)	Compulsory	One IPython Notebook file (.ipynb)	[student_number]-part-1.ipynb
Part 2 Notebook (Using the template provided on Learning Central)	Compulsory	One IPython Notebook file (.ipynb)	[student_number]-part-2.ipynb
Part 3 Notebook (Using the template provided on Learning Central)	Compulsory	One IPython Notebook file (.ipynb)	[student_number]-part-3.ipynb

If you are unable to submit your work, email your submission to the module leader and comsc-submissions@cardiff.ac.uk. If the technical issue is with Cardiff University systems, also report it to the University IT Support: email it-support@cardiff.ac.uk, tel. [+44 \(0\)29 2251 1111](tel:+4411292251111).

Assessment Description

You are tasked with analysing datasets representing different types of social and communication behaviour. These datasets are provided as files and can be found alongside this coursework pro-forma on Learning Central. Alongside the dataset files, there are 3 (THREE) IPython notebooks, named part-1.ipynb, part-2.ipynb, and part-3.ipynb, which you should solely use to complete the assignment and submit these in line with the Submission Instructions section above. The cells in each completed notebook will be ran in the order that they appear. You do not need to resubmit the dataset files.

You are required to address 16 total questions across the 3 parts. These questions are also listed below for convenience. For EACH question in EACH notebook:

1. Complete the cell below each question marked with “#CODE:” with the Python code needed to generate any new information you need for your answer. This information should be outputted when the cell is run, and any floating-point values should be presented to 2 decimal places unless they are less than 0.01.
2. Complete the cell below this marked with “ANSWER:” with your answer to the question, referring to the information outputted above (as well as any previous cell if needed). In doing so, briefly explain your approach and methods/measures used to answer the question and justify any choices made. Each answer cell should (ideally) be no more than 125 words.

You may use any Python packages installable via pip. “%pip install <some_package>” commands should be placed in the cell below “Install Python packages (pip only)” provided at the top of each notebook.

“import <some_package>” lines for all packages required for the notebook to be ran successfully should be placed in the cell under “Import Python packages” provided at the top of each notebook.

You may add additional cells throughout the notebooks, but this should be minimised.

Any code submitted will be run on a system equivalent to a Cardiff University imaged lab machine and must be submitted as stipulated in the instructions above. Any deviation from the submission instructions above (including the number and types of files submitted) may result in a mark of zero for the assessment or question part.

Staff reserve the right to invite students to a meeting to discuss coursework submissions.

Questions (duplicated from the three notebook files)

Part 1: Video Sharing Platform Interaction Behaviour

Examine the Graph Modelling Language (gml) files "social_network.gml" (social network) and "comment_network.gml" (comment network) which represent data for a sample of users on an online video sharing platform. Both networks are directed and share the same ids for nodes (anonymised users). However, the shared user ids are contained within the "label" attribute in the .gml files, not the node "id" attribute. Assume that all users have posted at least one video on the platform.

In the social network, an edge from a node, u , to some other node, v , indicates that u subscribes to v and their video content on the platform.

In the comment network, an edge from a node, u , to some other node, v , indicates that u posted a comment to one or more videos posted made by v . Edges are weighted with the weight representing the number of times this happened over the time period the dataset represents.

Using these networks, answer the following questions:

- Q1. How does the topological structure of the comment network differ from the social network in terms of the overall sparsity of edges between users and the number of connected groups of users?
- Q2. Do users typically only comment on other users' videos yet do not have any on their own videos, only receive comments by others on their videos but do not comment on others, both comment and receive comments, or neither comment nor receive comments?
- Q3. How many users only comment on videos where they also subscribe to the user who posted the video?
- Q4. How many users have only mutual subscription connections (i.e., every user that they subscribe to also subscribes to them) and only mutual comment connections with these same users?
- Q5. Do the 25 users that subscribe to the most other users also comment the most on videos by other users?
- Q6. To what extent does the number of subscribers a user has in the social network correlate with the number of users that have commented on their videos?

Part 2: Organisation "NotTeams" Messages

Examine the file "chat_data.edgelist" which represents instant messaging behaviour at an organisation. Each line contains two numbers, u and v , separated by a blank space. Consider each number as an identifier for an individual in an organisation, with the space on each line representing that the individual, u , sent at least one message to another individual, v , at some point. Assume that messages can only sent between pairs of users and therefore group messages are not possible.

Additionally, examine the JSON file "chat_data_departments.json" (departments file). Keys in the departments file represent individuals using the same ids as in the "chat_data.edgelist" file and the values represent a department id that the individual can be attributed to.

Model the data using an appropriate, directed network representation and answer the following questions:

- Q1. Could the connectivity within the network's largest, strongly connected component be suggested to be reflective of a small world phenomenon in comparison to the typical connectivity of 10 comparative random networks?
- Q2. How many individuals in the network have a higher or lower ratio of mutual connections than the ratio of mutual connections found in the overall network?
- Q3. Are occurrences of induced, connected subgraphs of 3 individuals (triads) with only mutual connections more abundant in the network than those with a mixture of mutual and asymmetric edges?
- Q4. How many individuals only send emails within their department?
- Q5. Are all departments with 10 or more members more tightly connected amongst themselves in comparison to all individuals across the overall network irrespective of their department? Where in this context, 'more tightly connected' is defined as having less connection sparsity and more reciprocated connections. In addition to answering the overall question as yes or no, provide a list of departments this is true for (if any) and not true for (if any).

Part 3: Email Data

Examine the file "email_data.csv" which represents email behaviour between individuals. Assume that emails are to one recipient only. Each row has three columns, representing a single event where a person (person_a) emailed another person (person_b) on some date and time of day (timestamp).

From this, answer the following questions:

- Q1. Build a suitable network to represent social connections based on the email behaviour that took place up to and including the first day of May. In doing so, assume that one or more emails from one person to another represents a mutual underlying social connection (i.e., regardless of whether person_a emailed person_b, vice versa, or both).

What non-existent edges would need to be added to the network to connect it so that a hypothetical message could pass along a path from any individual to any other? In doing so aim to minimise the longest shortest path as a result. Do not add these edges to the network for the later questions.

- Q2. Build a suitable network to represent social connections based on the cumulation of all email behaviour in the dataset. In doing so, assume that one or emails from one person to another represents a mutual underlying social connection (i.e., regardless of whether person_a emailed person_b, vice versa, or both).

Can the social phenomenon, 'Triadic Closure', be supported for the set of individuals that exist in both the network created from data up to and including the first day of May (i.e., from Q1) and the network built from all data?

- Q3. Using the largest connected components from both networks, is the average and median maximum degree of separation between individuals and all others larger in the network built from all data (i.e., from Q2) than the network built from data up to and including the first day of May (i.e., from Q1)?
- Q4. Using the largest connected component of network constructed from all data (i.e., from Q2), assume the role of an outsider with complete visibility of the network that now wishes to spread a hypothetical email such that everyone in the component would know the information it contained as quickly as possible. Assume that the information will spread in sequential timesteps using the following mechanism. If an individual is told the information in an email at timestep t , the individual will forward the email to all of their direct connections at timestep $t+1$. Individuals can therefore be told the information more than once.

If you could only select 1 individual to tell at timestep 0, what individuals could you select which would result in the message being received by everyone in the fewest timesteps as possible and what would the number of timesteps be?

- Q5. If you had to select any 5 individuals to tell at timestep 0, is there an example set of individuals that would result in the information being received by everyone in fewer timesteps than the single individual selection in Q4? In determining your answer, use one or more appropriate network connectivity measures, rather than an exhaustive search through every combination of individuals.

Assessment Criteria

Credit will be awarded against the following criteria. There are 100 marks available for this assignment. Each of the 16 questions are worth 6 marks, split between up to 3 marks for the approach and implementation and up to 3 marks for the explanations and justifications of the approach and implementation. This totals 96/100 possible marks. Marks will be awarded using the following criteria:

0 marks	1 mark	2 marks	3 marks
Unsuitable implementation that does not address the question. OR Non completion of the question.	Partially completed implementation that uses some appropriate selection of appropriate methods and measures.	Completed implementation with mostly appropriate selection and implementation of appropriate methods and measures.	Complete implementation with appropriate selection and implementation of methods and measures.
0 marks	1 mark	2 marks	3 marks
Little to no explanation of the approach taken in the implementation. OR The explanation is incorrect. OR	Partially incorrect description/explanation of the approach taken in the implementation. OR A brief description of the overall approach used in the implementation, but with missing or limited explanations of the	Some explanation of approach taken in the implementation with an explanation of why the specific methods or measures were selected in the implementation,	A clear, concise explanation and justification of the approach taken, including comparison against alternative, and potentially worse, choices of the methods or measures used where relevant.

Non completion of the question.	why the specific methods or measures were used.	but little to no explanation for why they are the most appropriate choice.	
---------------------------------	---	--	--

A further 4 marks (4/100 possible marks) will be awarded for the overall usability and readability of the notebooks, using the following criteria:

0 marks	1-2 marks	3-4 marks
No notebooks are runnable without modification due to errors.	All cells in some notebooks are runnable without modification due to errors. Some or most cells are clearly formatted without excessive commented out code and white space.	All cells in all notebooks are runnable without modification due to errors. Most or all cells are clearly formatted without excessive commented out code and white space. Floating point values are presented to 2 decimal places unless the value is less than 0.01.

The total mark awarded for this assessment aligns with the percentage boundaries for the following levels of attainment:

High Distinction 80%+	80 - 100 marks from the sum of marks for all 16 questions plus the overall usability and readability mark
Distinction 70-79%	70 - 79 marks from the sum of marks for all 16 questions plus the overall usability and readability mark
Merit 60-69%	60 - 69 marks from the sum of marks for all 16 questions plus the overall usability and readability mark
Pass 50-59%	50 - 59 marks from the sum of marks for all 16 questions plus the overall usability and readability mark
Marginal Fail 40-49%	40 - 49 marks from the sum of marks for all 16 questions plus the overall usability and readability mark
Fail 0-39%	0 - 39 marks from the sum of marks for all 16 questions plus the overall usability and readability mark

Help and Support

Questions about the assessment can be asked at the beginning or end of the synchronous sessions in Weeks 6-10.

Feedback

Feedback on your coursework will address the above assessment criteria. Feedback and marks will be returned via Learning Central in accordance with the university's [Academic Feedback Policy](#). You can see feedback return dates for each assessment on Learning Central: see '[COMSC Assessment Calendar 2025 - 26](#)' page.